Free State AG Solar System Report

Energy Analysis and Financial Overview

Location

Zürich Hardturm, Zürich, Switzerland

Customer

Test Test

433	194.9	256'540
panels	kWp	kWh/Year

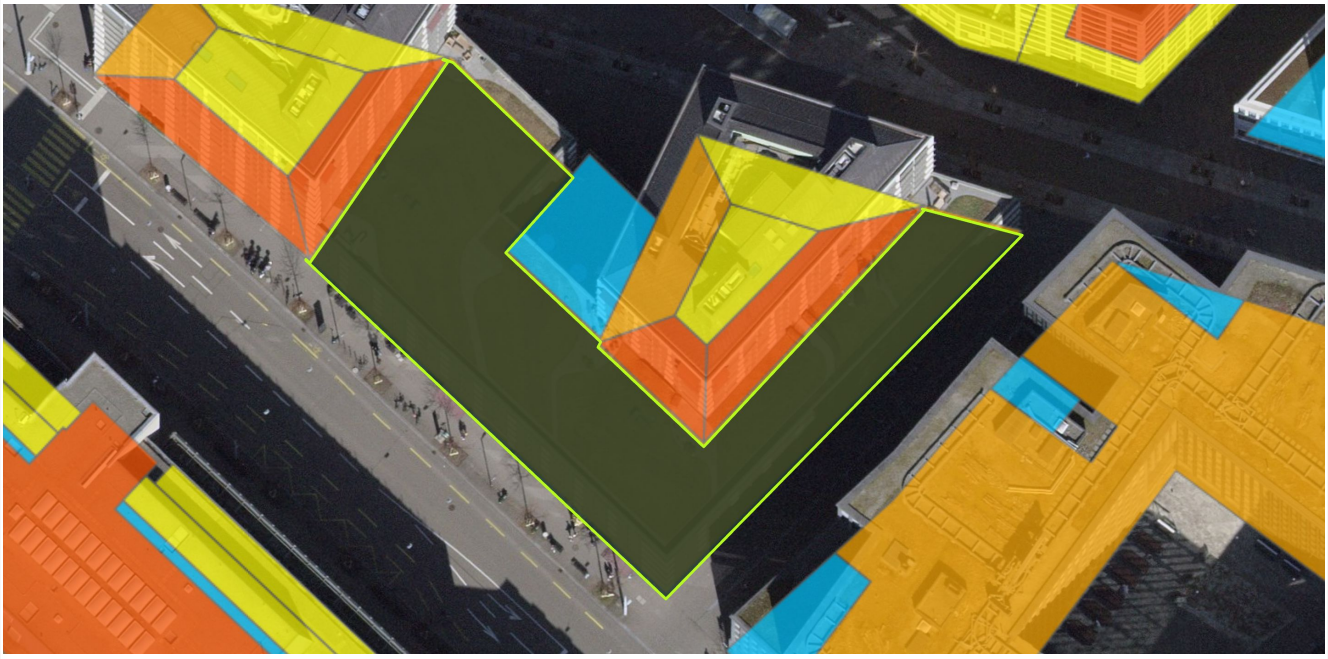
CO2 Savings

32'837 kg/per year

Contents

Roof View with Solar Panels	3
Energy Flow	4
Monthly Energy Production (kWh)	5
Monthly Production Details	6
Environmental Impact	7
Equipment Overview	8
Financial Summary	9
Next Steps	10

Roof View with Solar Panels



System Details

Number of Panels	**433 panels**	Yearly Production	**256'540 kWh**
System Power	**194.85 kWp**	Daily Average	**703.0 kWh**
Orientation	**-180° (N)**	Electricity Price	**27.7 Rp/kWh**
Tilt	**0°**	Feed-in Tariff	**8.0 Rp/kWh**

Energy Flow

Yearly Production

256'540 kWh

Consumption

15'550 kWh

Own Consumption

71'581 kWh

Independence

460%

Grid Supply

0 kWh

Feed-in to Grid

184'959 kWh

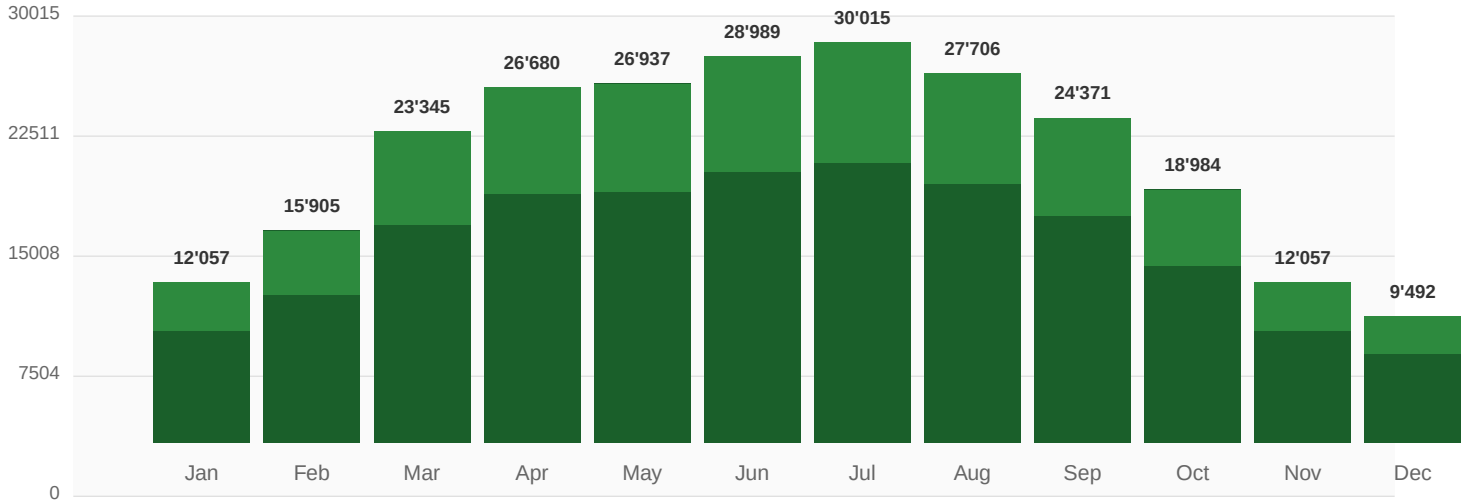
Self Consumption

28%

Cost Comparison

Grid electricity price	**27.7 Rp/kWh**
Solar electricity cost	**4.1 Rp/kWh**
Cost advantage	**85%**

Monthly Energy Production (kWh)



Energy Distribution

Self Consumption: 71'581 kWh

Grid Export: 184'959 kWh

Annual Savings

Self Consumption	71'581 kWh x 27.7 Rp	**CHF 19'828**
Grid Export	184'959 kWh x 8.0 Rp	**CHF 14'797**
Total Annual Savings		**CHF 19'828**

Data source: Sonnendach.ch - Swiss Federal Office of Energy

Monthly Production Details

	Consumption	Yearly	Self Con%	Own Consumption	Feed-in to Grid
Jan	1'555 kWh	12'057 kWh	13%	1'555 kWh	10'502 kWh
Feb	1'400 kWh	15'905 kWh	9%	1'400 kWh	14'505 kWh
Mar	1'400 kWh	23'345 kWh	6%	1'400 kWh	21'945 kWh
Apr	1'244 kWh	26'680 kWh	5%	1'244 kWh	25'436 kWh
May	1'089 kWh	26'937 kWh	4%	1'089 kWh	25'848 kWh
Jun	1'089 kWh	28'989 kWh	4%	1'089 kWh	27'900 kWh
Jul	1'089 kWh	30'015 kWh	4%	1'089 kWh	28'926 kWh
Aug	1'089 kWh	27'706 kWh	4%	1'089 kWh	26'617 kWh
Sep	1'244 kWh	24'371 kWh	5%	1'244 kWh	23'127 kWh
Oct	1'400 kWh	18'984 kWh	7%	1'400 kWh	17'584 kWh
Nov	1'400 kWh	12'057 kWh	12%	1'400 kWh	10'657 kWh
Dec	1'555 kWh	9'492 kWh	16%	1'555 kWh	7'937 kWh
Total	**15'554 kWh**	**256'540 kWh**	**6%**	**15'554 kWh**	**240'984 kWh**

Environmental Impact

1'539'240

km

Electric car distance

can be driven with annual production

2

years

Grey energy payback

to produce the energy used for manufacturing

32'837

kg CO2

CO2 Savings

per year

Modern solar panels contain neither heavy metals nor rare earths and can be almost 100% recycled. Solar energy has an excellent eco-balance.

Data source: Sonnendach.ch - Swiss Federal Office of Energy

Equipment Overview

Solar Panel

JA Solar DeepBlue 4.0 Pro 450W

Inverter

Huawei SUN2000 5kW

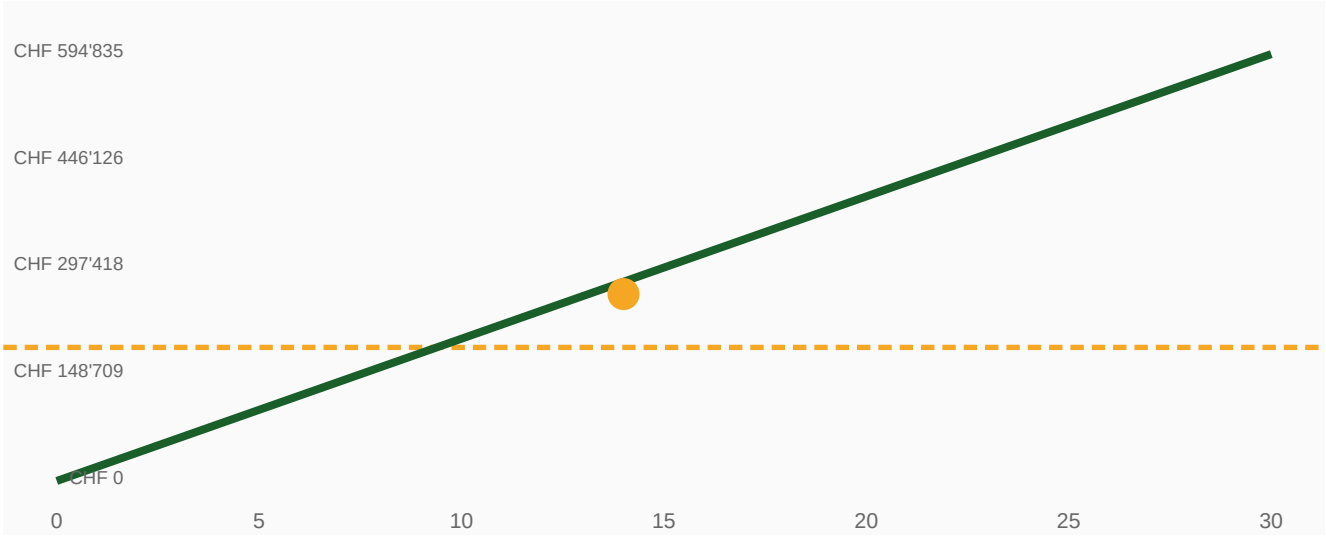
Battery Storage

BYD BatteryBox HVS 10.2

Mounting System

K2 CrossRail 2.1

30-Year Financial Projection



Cumulative Savings | Investment (CHF 260'475) | Payback Point (14.0 years)

Financial Summary

Total Investment	**CHF 292'275**
Subsidies	**- CHF 31'800**
Net Investment	**CHF 260'475**
Annual Savings	**CHF 19'828**
30-Year Savings	**CHF 693'974**
Payback Period	**14.0 years**

Next Steps

This report provides you with a non-binding estimate of price, production, and profitability of a solar system on your building.

Contact us for a binding offer

Free State AG
info@freestate.ch | www.freestate.ch

The information in this report is non-binding and must be verified by a specialist.